

STEPS TO RUN:

STEP1: (Master and Slave)

- 1) `sudo apt update`
- 2) `sudo swapoff -a`
- 3) `sudo sed -i ' / swap / s/^\(.*\)$/#\1/g' /etc/fstab`
- 4) `sudo tee /etc/modules-load.d/containerd.conf <<EOF`
`overlay`
`br_netfilter`
`EOF`
- 5) `sudo modprobe overlay`
- 6) `sudo modprobe br_netfilter`
- 7) `sudo tee /etc/sysctl.d/kubernetes.conf <<EOF`
`net.bridge.bridge-nf-call-ip6tables = 1`
`net.bridge.bridge-nf-call-iptables = 1`
`net.ipv4.ip_forward = 1`
`EOF`
- 8) `sudo sysctl --system`
- 9) `sudo apt install ca-certificates curl gnupg lsb-release`
- 10) `sudo mkdir -p /etc/apt/keyrings`
- 11) `curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o /etc/apt/keyrings/docker.gpg`
- 12) `echo "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg] https://download.docker.com/linux/ubuntu $(lsb_release -cs) stable" | sudo tee /etc/apt/sources.list.d/docker.list > /dev/null`
- 13) `sudo apt update && sudo apt install -y containerd.io`
- 14) `containerd config default | sudo tee /etc/containerd/config.toml >/dev/null 2>&1`
- 15) `sudo sed -i 's/SystemdCgroup \= false/SystemdCgroup \= true/g' /etc/containerd/config.toml`
- 16) `sudo systemctl restart containerd`
- 17) `sudo systemctl enable containerd`

STEP 2: (Master and Slave)

- 1) `sudo apt-get update`
- 2) `sudo apt-get install -y apt-transport-https ca-certificates curl gpg`
- 3) `curl -fsSL https://pkgs.k8s.io/core:/stable:/v1.30/deb/Release.key | gpg --dearmor -o /etc/apt/keyrings/kubernetes-apt-keyring.gpg`
- 4) `echo 'deb [signed-by=/etc/apt/keyrings/kubernetes-apt-keyring.gpg] https://pkgs.k8s.io/core:/stable:/v1.30/deb/ /' | tee /etc/apt/sources.list.d/kubernetes.list`
- 5) `apt update`
- 6) `apt-get update`

7) apt-get install -y kubelet kubeadm kubectl

8) apt-mark hold kubelet kubeadm kubectl

9) sudo systemctl enable --now kubelet

STEP3: (Only on Master)

1) kubeadm init --pod-network-cidr=10.244.0.0/16

2) kubectl apply -f <https://docs.projectcalico.org/manifests/calico.yaml>

3) mkdir -p \$HOME/.kube

4) sudo cp -i /etc/kubernetes/admin.conf \$HOME/.kube/config

5) sudo chown \$(id -u):\$(id -g) \$HOME/.kube/config

6) export KUBECONFIG=/etc/kubernetes/admin.conf

7) kubectl get nodes

8) kubectl apply -f <https://docs.projectcalico.org/manifests/calico.yaml>

9) kubectl get pods -A

10) kubectl get nodes

#To get the token which gets generated after (kubeadm init --pod-network-cidr=10.244.0.0/16) we can execute the below command:

kubeadm token create --print-join-command